

Response Letter with Annotations

Krause, N. M., Freiling, I., & Scheufele, D. A. (2025). Our changing information ecosystem for science and why it matters for effective science communication. *Proceedings of the National Academy of Sciences*, 122(27), e2400928121. <https://doi.org/10.1073/pnas.2400928121>

Annotations

1. Purposes

- a. Motivation: Scientific organizations' struggles to adapt to evolving media are linked, in part, to asymmetries in which social media platforms prevent researchers from producing reliable data that could inform institutional change and improve science communication
- b. Specific Rationale
 - Emerging online platforms change how science is communicated. Specifically, traditional science journalism and institutions need to compete for audience's attention with diverse entities, e.g., celebrity scientists, conspiracy theorists, across different platforms
 - It's necessary to learn current information ecologies to enact effective science communication, but lagging theoretical work and conceptual slippage, methodological limitations related to data access constraints, and imbalances of power in academia–industry collaborations impose great challenges
- c. Research Questions and/or Hypotheses: N/A
- d. Main idea
 - To understand the current information dynamics, researchers need to a) obtain access to proprietary data, b) reconceptualize information ecologies as social systems, c) avoid ceding core research tasks to platforms, d) address ethical issues, and e) grapple with the urgency of moving forward productively

2. Method

- a. Research Design
 - Critical review
- b. Case: COVID-19
 - COVID-19 pandemic is one of the issues that are likely to require difficult trade-offs across stakeholders whose values and priorities diverge, and it highlighted how ill-prepared or underresourced scientific institutions are in terms of effectively leveraging current media ecologies to support nimble risk mitigation initiatives or scalable democratic deliberation with respect to high-stakes, uncertain science
- c. Measures: N/A
- d. Analysis: N/A

3. Results and Conclusions

a. Current media ecologies of science communication

- Users are continuously multitasking in a multimodality environment, thus,
 1. science communicators cannot realistically capture end-to-end processes of content selection, consumption, sharing, and response on a single platform
 2. Studies of (science) communication in new information ecologies can expect to increasingly struggle with issues of ecological validity and the generalizability of findings across contexts, cultures, and time
- Platforms prioritize engagement and user retention, thus,
 1. science-related information will struggle to compete
 2. science communicators struggle to use emotion as well as personal narrative alongside facts in ways that keep public discourse about science “grounded in reality”
- Scientific “Authority” as gatekeeping evolves, specifically,
 1. Scientists must compete in an attention economy in which they are just one of many voices
 2. Proprietary algorithms now serve as a new kind of gatekeeper
- Information spill over from one context or platform to another, but
 1. It’s hard to track and control the spillover of messages
 2. Trace data is replete with ethical and practical limitations
- Integrating personal speech of scientists or scientific influencers in current media environments may result in negative attitude towards scientists, punitive actions against scientists, and make them involve in existing social conflicts

b. High-stakes and uncertain issues like COVID-19 add extra challenges to science communication, e.g.,

- The public and policymakers watched as scientists engaged in an accelerated version of the messy, self-correcting work. It’s necessary for scientific process, but in the context of COVID-19 and online environments, it can be interpreted as politicalized and as implicated in identity-based disputes
- Policy based on uncertain science may be one-sided and cause damage to society
- Scientists seem to show bias towards certain parties, thereby exacerbating party polarization

c. Overall challenges faced by science communication: lacking sufficiently clear data to discuss whether social media platforms are the most productive channel for convincing skeptical publics about science

- **Data accessibility: Researchers do not have consistent access to most user, platform, and content data (e.g., 2020 Meta Project)** As a compromise, they can only

1. simulate selective instances of online behavior or information exposure in traditional experimental or other behavioral science studies
 2. have limited access to carefully curated subsets of data through API or scraping
 3. collaborate with social media platforms, which sometimes means granting industry partners control over the coding of variables and the types of primary data that are shared or not shared, among other things
- Extending research questions from a single platform to a broader social system (e.g., other platforms, offline interaction)
 - Core research tasks cannot be ceded to platforms (e.g., the conceptualization and operationalization) – 2020 Meta project. It may also lead to further concern about COI
 - Ethical challenges:
 1. Collecting digital trace data without consent by participants – alternative: data donation. But it also has the problem of biased data due to self-selection
 2. Need to protect user privacy, especially when the exact wording of UGC will be shared
 3. Platform-controlled experiments or experiments that result from academic-platform collaborations can fail to minimize harm to the unwitting users of these platforms, violating the ethical research principle of beneficence, among other concerns
 - Construct productive paths forward (see “conclusions”)
- d. Authors’ Conclusions: possible solutions
- “good-faith” compromise between industry and academia: e.g., “gated” panels of users on various platforms; independent advisory boards that include independently and equitably appointed members representing a wide range of stakeholder groups
 - the National Academies of Sciences, Engineering, and Medicine (NASEM) should convene dialogues across academia, industry, and other public stakeholders to generate and execute solutions to the challenges we have described, some of which require reflexivity within science itself

4. Discussion

- a. N/A

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Response Letter (~600 words)

The study provides valid discussions about how the evolving media dynamic interact with high risk and uncertain events to impose challenges on effective science communication. Specifically, it identifies that science communication needs to compete for users' fragmented attention in an environment where there are many other voices, and science content is not prioritized. Direct communications of scientists may be perceived as politicalized. It's also hard to trace users' activities on multiple platforms and offline communication. All of those characteristics are furthered strengthened in COVID-19 related science communication. The paper also provides inspiring vision of treating information ecologies as social systems and keeping independence of academic researchers.

While I appreciate the discussion and authors' efforts to provide possible solutions to those challenges, I hold a relatively pessimistic idea of the feasibility of those solutions and have concern about the conceptualization of "science communication" in this paper. Firstly, in terms of the solutions, the authors call for increased data transparency and more "complete" data for researchers. While it is ideal, it might overlook the great power asymmetry between big tech companies and academia. For companies, business interests are prioritized and perhaps, outweigh academia interests. Since user data construct the competitive advantages of platforms, without legal requirement, I am pessimistic about the incentives for those companies to voluntarily share their data, especially when the academic research or findings may possibly misalign with the companies' interests.

Second, this paper seems to frame those challenges simply from the perspective of scientific institutions and scientists, and treat science communication in new media dynamics as a process of maintaining "authority". This seems to reinforce a "deficient model" of science communication, which emphasize that science communication is a one-way process of knowledge propagation from scientists to public. As a result, it may overlook the potential opportunities brought by publics' active engagement in science communication. For example, science influencers may do well in balancing serious scientific facts with appealing formats.

Third, when discussing the possible influence of platform design, apart from realizing its effects on polarized content and emotions, perhaps a more interesting concern would be: what are we truly observing on social media platforms? The platforms and embedded algorithms, affordances, are designed based on certain social science theories. So, while we use so-called behavioral data to analyze and successfully identify some kinds of patterns, are we just capturing the response of such design, the underlying principles of theories, or the human behavior itself?